

Math 432/532: Introduction to Topology II

Solutions to HW #4

1. Suppose that L is a simplicial complex and $K_1, K_2 \subset L$ are subcomplexes. Show that $\chi(K_1 \cup K_2) = \chi(K_1) + \chi(K_2) - \chi(K_1 \cap K_2)$.

Solution: This is essentially the inclusion-exclusion principle. For any i , the number of i -dimensional simplices in $K_1 \cup K_2$ is equal to the number in K_1 plus the number in K_2 minus the number in both (because these have been overcounted). Taking the alternating sum over all i gives the desired statement.

2. Suppose that K is a simplicial complex, and consider its barycentric subdivision K^1 . It is true in general that $\chi(K^1) = \chi(K)$, but the proof involves more combinatorial bookkeeping than I want to ask you to do on a homework assignment. Instead, I'm just going to ask you to prove it for complexes of dimension at most 2.

(i) Prove it when $\dim K = 1$.

Solution: Let α_i be the number of i -simplices of K , and β_i the number of i -simplices of K^1 . Then $\beta_0 = \alpha_0 + \alpha_1$ and $\beta_1 = 2\alpha_1$, so $\chi(K^1) = \beta_0 - \beta_1 = \alpha_0 + \alpha_1 - 2\alpha_1 = \alpha_0 - \alpha_1 = \chi(K)$.

(ii) Prove it when $\dim K = 2$.

Solution: Let α_i be the number of i -simplices of K , and β_i the number of i -simplices of K^1 . Then $\beta_0 = \alpha_0 + \alpha_1 + \alpha_2$, $\beta_1 = 2\alpha_1 + 6\alpha_2$, and $\beta_2 = 6\alpha_2$, so

$$\chi(K^1) = \beta_0 - \beta_1 + \beta_2 = \alpha_0 + \alpha_1 + \alpha_2 - 2\alpha_1 - 6\alpha_2 + 6\alpha_2 = \alpha_0 - \alpha_1 + \alpha_2 = \chi(K).$$

3. Let X be the Klein bottle, and consider the three simple closed curves shown in Figure 7.16 of your book. For each of these curves, determine whether it thickens to a cylinder or a Möbius strip, and describe the surface obtained by a single surgery. (Note that you can answer this question without thinking about triangulations.)

Solution: The curve indicated by the vertical line thickens to a Möbius strip, and surgery along this curve gives a surface homeomorphic to $\mathbb{R}P^2$. Same with the diagonal line. The curve indicated by the horizontal line thickens to a cylinder, and surgery along this curve gives a surface homeomorphic to S^2 . In this case, undoing the surgery involves adding a handle in the funny way shown in Figure 7.2(b).